

Tannins

DR. Murtadha saleh

❑ **Tannins** compromise a large group of complex substances that are widely distributed in the plant kingdom.

❑ Tannins precipitate **proteins**

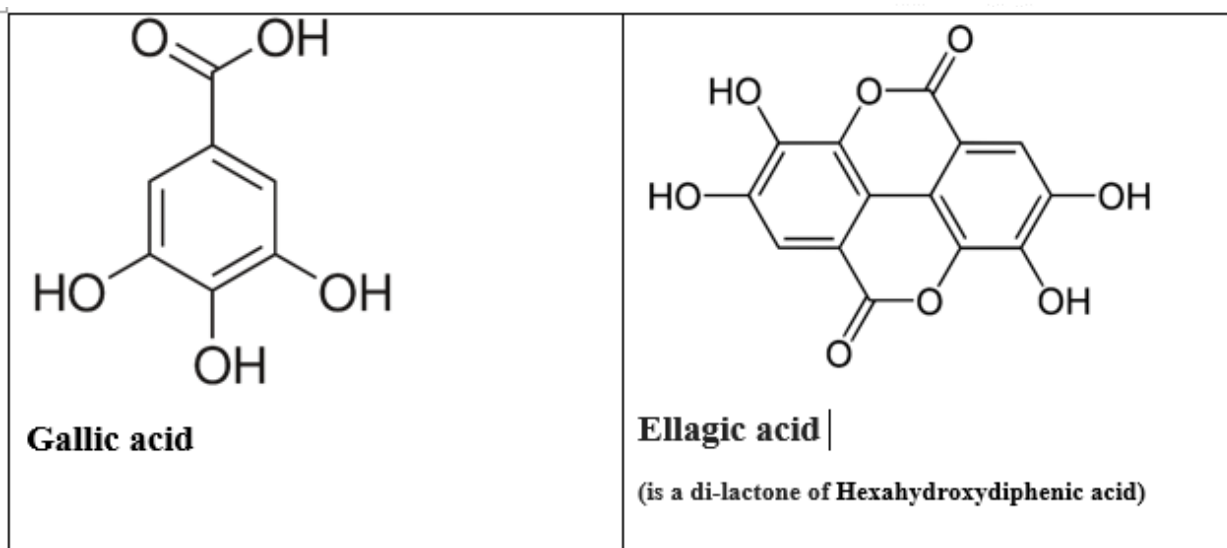
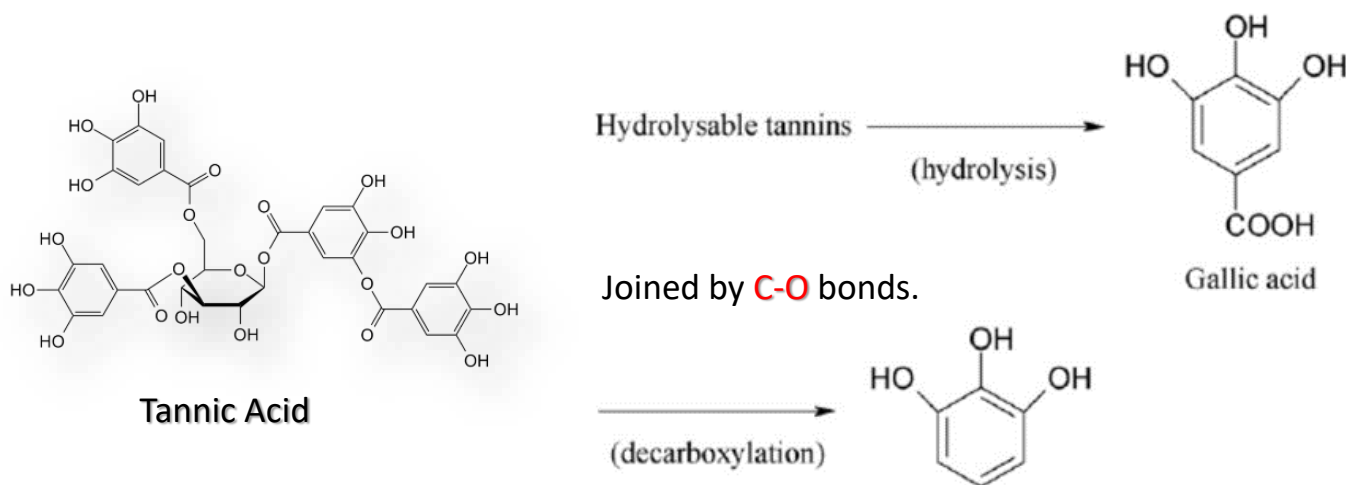
❑ When applied to living tissue this action is known as an "**astringent**"

Chemistry

❑ Poly hydroxyl phenols

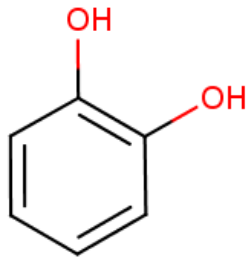
❑ Tannins are divided according to the identity of the phenolic nuclei involved, and on the way they are joined into 2 classes:

1-Hydrolysable tannins:

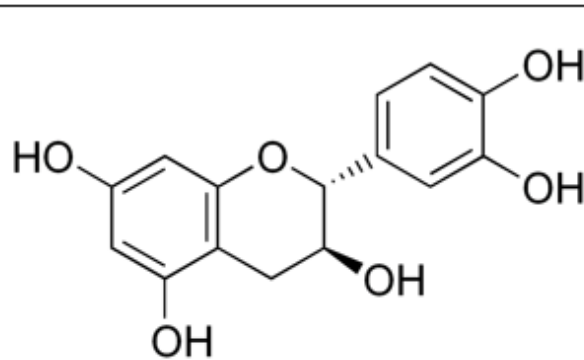
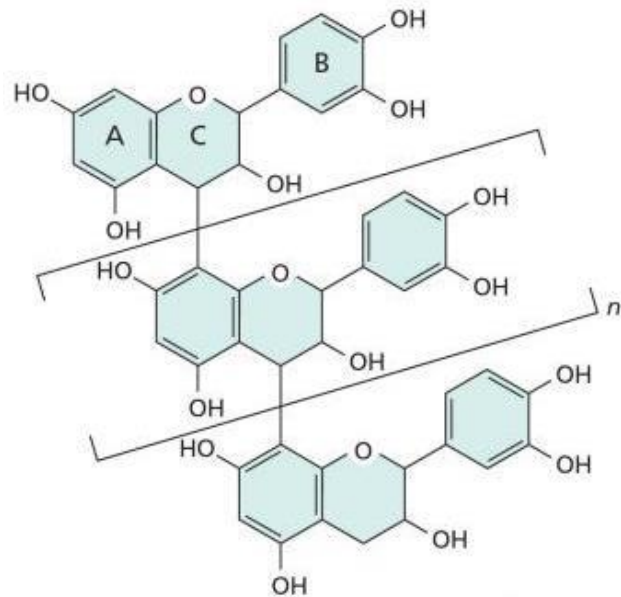


2-Non-hydrolysable tannins

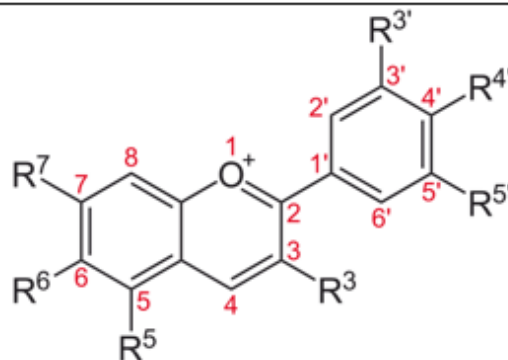
► Polymers, Joined by **C-C** bonds.



Catechol



Catechin



Anthocyanins

(Are glycosides of anthocyanidins)

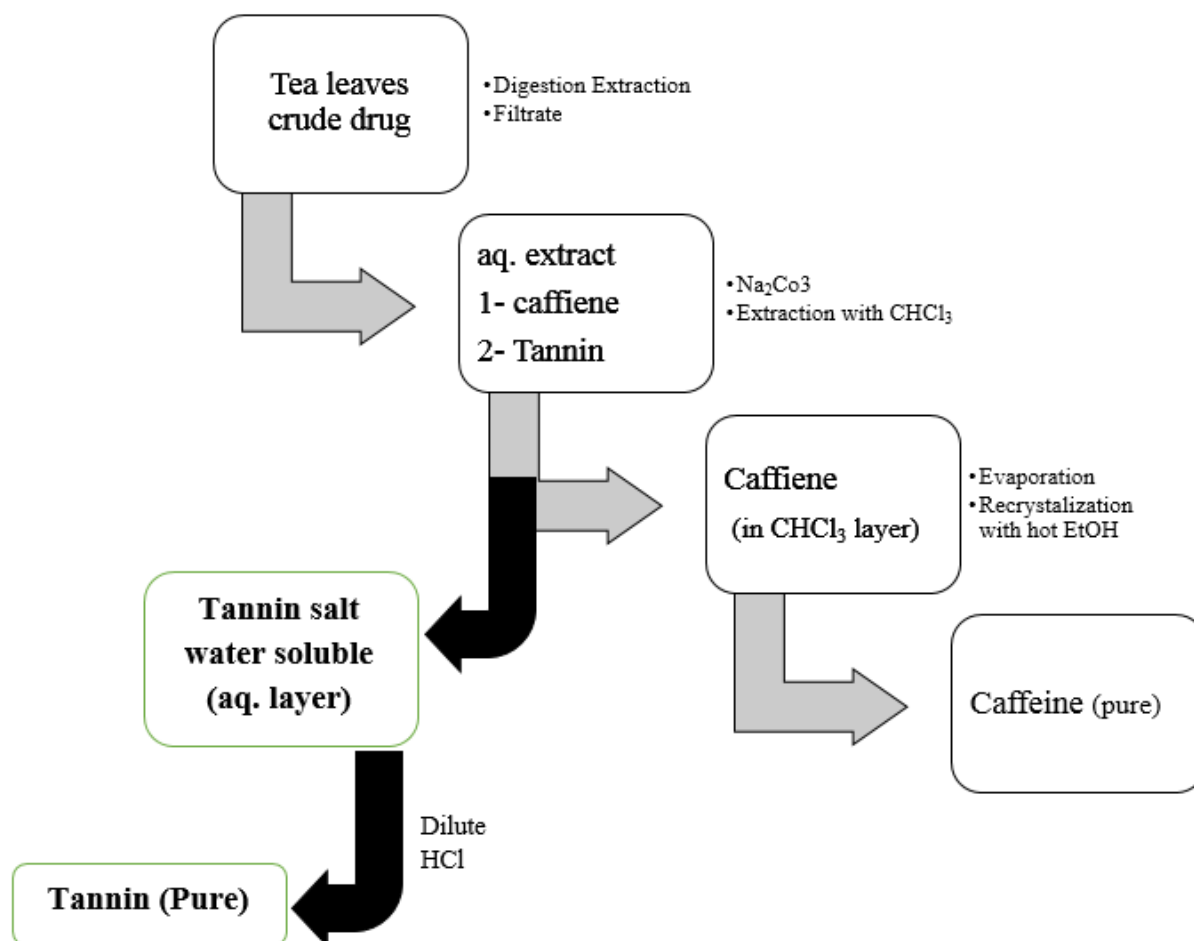
Uses of tannins

1) **Astringents**, used in the gastrointestinal tract and on the skin abrasion.

2) **In the treatment of burns**, the proteins of the exposed tissue are precipitated and form a mildly antiseptic protective coat under which the regeneration of new tissue may take place.

3) **Antidote** treatment of alkaloids poisoning.

Extraction & identification of Tannins:



Identification of tannins:

☞ To aqueous solution (1ml), 2 drops of 5% FeCl₃

solution is added, a **dark green** or **blue** is formed depending on the type of the present tannin. In case of catechol tannin (found in Tea) a dark green is resulted.